

Behaving Sustainability: Cultural Cognition, Environmental Orientation, and Pro-Environmental Behavior

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Introduction

Achieving environmentally sustainable outcomes likely requires some behavioral changes at the individual, community, and societal levels. However, not all individuals are equally likely to engage in pro-environmental behaviors, and understanding the psychological and cultural factors that drive these behaviors is important for enhancing sustainability. Environmental behaviors are widely studied (Zhu, Tong, and Song 2021), and scholars of environmental psychology often lament the so-called ‘attitude-behavior’ gap wherein environmental attitudes do not explain very much of the variance in environmental behavior (Gifford and Nilsson 2014; Kollmuss and Agyeman 2002; Stern 2000). If people care strongly about the environment, why don’t they act in a more sustainable way? The attitude-behavior gap is not because people are hypocrites, but because researchers do not yet have a complete attitude-behavior model (Kaiser, Kibbe, and Hentschke 2021).

To understand how environmental attitudes influence behaviors, scholars often use approaches such as the Value-Belief-Norm (VBN) framework, where higher-order values shape second-order beliefs that, in turn, influence behaviors (Canlas, Karpudewan, and Khan 2022; Stern et al. 1999; Stern 2000; Vysevkova et al. 2026). Higher-order values form a cognitive foundation that influences how information about the world is processed and understood across multiple issues (Ripberger et al. 2014). When considering environmental behaviors, the second-order values involve environmental cognition, such as views about the relationship of humans to the natural world and how information about the environment is understood (Henry and Dietz 2012; Stern and Dietz 1994). In addition to environmental cognition, there are also affective feelings that individuals hold about the environment (Gkargkavouzi, Halkos, and Matsiori 2019; Mayer and Frantz 2004). Together environmental cognition and affective feelings form an environmental orientation that shapes environmental behaviors. However, to develop a complete attitude-behavior model, it is important to understand all of the various links in the chain, including the higher-order values and cognition that inform environmental

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orientations. Of note, the cultural worldviews measured by cultural cognition have been shown to shape environmental orientations (Nowlin 2019).

Cultural cognition was developed as a tool to measure how cultural values shape beliefs about risk (e.g., Kahan et al. 2007). It is based on cultural theory and its conceptualization of worldviews (Douglas and Wildavsky 1982; Kahan 2012). Cultural theory addresses how cultural values influence how people perceive and respond to risk. By measuring the cultural values of an individual, we may be able to predict how they will perceive and respond to risks associated with certain activities or products. We are interested in how cultural cognition shapes views toward environmental issues and how those views shape behavior. Cultural worldviews their subsequent risk perceptions are likely strong drivers of environmental orientation because of the nature of environmental problems typically involves elements of risk or harm to the environment.

The cultural worldviews associated with cultural cognition are measured in two dimensions, a grid dimension (egalitarianism-hierarchy) and a group dimension (communitarianism-individualism). In general, those that are more egalitarian and communitarian are more concerned about environmental risks than those that are more hierarchical and individualist. Therefore, it is expected that individuals high in egalitarianism and communitarianism should engage in pro-environmental behaviors to mitigate risks to the natural environment. However, previous research shows only a weak correlation between egalitarianism and communitarianism and pro-environmental behavior (Sparks et al. 2021). Therefore, cultural cognition shapes how individuals think and process information about issues, but may not directly influence behaviors.

In this paper, we develop a conceptual framework that accounts for both the cognitive and affective influences on pro-environmental behavior. Specifically, we posit that that higher-order beliefs, drawn from cultural cognition (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006), inform environmental orientations (Henry and Dietz 2012), which in turn influence behavior, both public and private. Drawing from Henry and Dietz (2012), we use the New Ecological Paradigm (NEP) as our measure of environmental cognition, and the Connectedness to Nature Scale (CNS) as the affective element of environmental orientation. Previous work failed to find that cultural cognition influence environmental behavior; however, it has been found to influence environmental cognition using the NEP (Nowlin 2019). Therefore, we hypothesize that environmental orientations mediate the relationship between cultural cognition and behavior.

Overall, we find egalitarian communitarians hold an ecological worldview and feel more connected to nature. However, hierarchical individualists are less concern about the environment than other worldviews. Additionally, we find that ecological worldviews are associated with feeling connected to nature, and that both worldviews and affective nature connection are associated with increase in self-reported pro-environmental behaviors. Finally, we find that environmental orientations mediate the relationship between cultural worldviews and behaviors, meaning that cultural worldviews play an important, but indirect, role in environmental behaviors.

Cognition, Environmental Orientations, and Pro-Environmental Behavior

The process though values and beliefs impact behavior is often understood as a process where higher-order, antecedent values influence second-order beliefs, which in turn influence behavior (Dietz, Fitzgerald, and

Shwom 2005; Steg et al. 2011). One of the leading frameworks that adopts this approach is the Value-Belief-Norm (VBN) framework (Stern et al. 1999; Stern 2000). In brief, the VBN posits that underlining values inform broad environmental orientations that activate norms that in turn support pro-environmental behavior. VBN involves a sequential chain that goes from values to beliefs to behavior. Therefore, values inform attitudes, which in turn inform behaviors.

As noted, the VBN framework moves from values to beliefs to norms to pro-environmental behaviors. The values that have been identified and applied include biospheric, altruistic, and egoistic (Canlas, Karpudewan, and Khan 2022; Stern 2000). Biospheric values are associated with concern for the environment, altruistic values involved concern for others, and egoistic values involve concern for self (Fornara et al. 2020). In general, biospheric and altruistic values are associated with an ecological worldview, whereas egoistic values are not. Beliefs are grounded in ecological worldviews, which are typically operationalized using the New Ecological Paradigm (NEP) measure (Chen 2015; Fornara et al. 2020; Slimak and Dietz 2006; Stern et al. 1999). Additional beliefs include an awareness of consequences (AC) and an ascription of responsibility (AR), which active norms involving a sense of obligation to action (Stern 2000). The behaviors identified include private consumption-based behaviors, public facing activism, and support for environmental policies (Stern et al. 1999). VBN is widely used (Canlas, Karpudewan, and Khan 2022; Vysevkova et al. 2026) across multiple contexts, including climate mitigation behaviors in Taiwan (Chen 2015), energy conservation (Yeboah and Kaplowitz 2016), biodiversity in Europe (Fornara et al. 2020; Riepe et al. 2021), and perceptions of environmental risk (Slimak and Dietz 2006).

A similar set of assumptions inform an understanding of belief systems developed by policy scholars using the Advocacy Coalition Framework (ACF) (Henry and Dietz 2012; Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). With this approach, beliefs systems are interrelated beliefs made up of three different types, including deep core, policy core, and secondary beliefs. Deep core beliefs are the broadest in scope and are fundamental beliefs that inform attitudes about multiple issues. They tend to be stable, and play a role in how issues are understood and they influence reasoning about issues (Kahan et al. 2012; Nohrstedt et al. 2023). Biospheric, altruistic, and egoistic values, could seen as examples of deep core beliefs, as could political ideology. Next, are policy core beliefs, which are beliefs about particular issues. Environmental orientations would be akin to policy core beliefs. Finally, secondary aspects are the narrowest in scope and include beliefs about how best to achieve the goals outlined in policy core beliefs. We aim to develop previous work that noted a connection between VBN and the ACF (Henry and Dietz 2012).

In developing our approach, we draw from both VBN and the ACF by positing a sequential chain and a hierarchical ordering of values, beliefs, and attitudes that lead from abstract values to concrete actions. Specifically, we note that deep core beliefs inform environmental orientations that, in turn, make pro-environmental behaviors more (or less) likely. In addition, we separate beliefs that shape *cognition* from the affective and identity elements of environmental orientations. We note that the role of cognition has not been well developed in approaches to understanding environmental behaviors, despite the importance that the way worldviews shape information processing has on midstream beliefs and downstream behaviors (Liobikienė and Poškus 2019; Ripberger et al. 2014). Our measure of deep core beliefs is cultural cognition, which as the name implies, involve deeply rooted values and beliefs that influence cognition. For environmental orientations, we draw on environmental cognition, which is “the way individuals structure their thinking about environmental issues and associated political actions” (Henry and Dietz 2012, 238). We argue that environmental orientations

contain a cognitive element as well as an affective element. Deep core beliefs influence both the cognitive and affective aspects of environmental orientations. Finally, deep core beliefs and environmental orientations influence pro-environmental behaviors; however, it is not clear if deep core beliefs influence behaviors directly (Markle 2019) or indirectly (Sparks et al. 2021). However, the affective element of environmental orientations tends to have the most direct influence on behaviors (Mackay and Schmitt 2019; Sparks et al. 2021).

Deep Core Beliefs: Cultural Cognition

For our measure of deep core beliefs we use cultural cognition. Cultural cognition is derived from the cultural theory of risk developed by Mary Douglas and Aaron Wildavsky (Douglas and Wildavsky 1982; Kahan 2012; Thompson, Ellis, and Wildavsky 1990). Cultural theory (CT) has been identified as a leading way to conceptualize and measure deep core beliefs (Ripberger et al. 2014). In brief, CT posits that social structures and relationships are represented by two dimensions, including a grid dimension and a group dimension. Four competing cultural worldviews are possible based on positioning in the grid and group dimensional space. The grid dimension represents preferred social structures regarding the roles that one plays in society and the rules by which one must abide. The group dimension represents the importance of cohesive social units and the well-being of the broader community. The original formulation of CT produces four cultural worldviews based on the relative positions on the grid and group dimensions, including hierarchy (high grid, high group), egalitarianism (low grid, high group), individualism (low grid, and low group), and fatalism (high grid, low group).

Cultural worldviews provide a foundational set of deep core beliefs that inform beliefs about specific policy issues (i.e., policy core beliefs), societal controversies, and risks, among other beliefs (Jenkins-Smith et al. 2014; Nowlin 2019; Ripberger et al. 2014; Sotirov and Winkel 2016). In the context of cultural worldviews, perceptions of risk are based on the degree to which they undermine or threaten the preferred social order within each worldview. Several studies have shown a connection between cultural worldviews and risk perceptions associated with a host of environmental issues (see Xue et al. 2014), including fracking (Boudet et al. 2014), nuclear waste (Jenkins-Smith and Smith 1994), renewable energy (Tumilson et al. 2018), geoengineering (Kahan et al. 2015), natural resource management (Quan, Song, and Nguyen 2025), high-voltage power lines (Moyer and Song 2019), and climate change (Jones 2011; Kahan et al. 2012; Nowlin 2022; Nowlin and Rabovsky 2020), among others.

Cultural cognition is based on CT and its cultural worldviews, but makes two important distinctions. The first is its focus on social and psychological mechanisms that underlie how worldviews influence opinion on contested political and policy issues. Cultural commitments occur prior to facts, “in the cognitive sense that what citizens believe about the empirical consequences of [controversial] policies *derives* from their cultural worldviews” (Kahan and Braman 2006, 148). Individuals rely on five mechanisms to ensure that their positions on issues are congruent with their cultural worldviews. These mechanisms include, 1) *cultural identity-protective cognition*, where individuals conform their beliefs to those in their “group” that share cultural values; 2) *culturally biased assimilation and cultural polarization*, involves selective attention to worldview-confirming information and balanced information tends drives people of opposing worldviews further apart rather than together; 3) *cultural availability* occurs as worldview-confirming events and experts are more readily noticed, credited, and recalled; 4) *cultural credibility heuristic* is when people trust sources

whose apparent values match their own; and 5) *cultural-identity affirmation*, has been shown when framing information alongside an identity-affirming rather than identity-threatening policy solution makes people more receptive to the underlying facts (Kahan 2012). Together, these mechanisms account for the cognition part of cultural cognition.

The second distinction of cultural cognition from CT is how cultural worldviews are measured empirically. As noted, the CT worldviews are based on the placement on the grid and group dimensions, and worldviews are a result of placement on both. However, cultural cognition conceives grid and group as two orthogonal dimensions with worldviews identified as being high or low on each dimension rather than a combination of the dimensions. Those high on the grid dimension are called hierarchical as they prefer a society with well-defined roles and clear lines of authority, whereas those low on the grid dimension are termed egalitarian as they value a social order based on an equal sharing of power and resources. Communitarian is the worldview that is high on the group dimension, and it represents a preference for a collectivist society. Finally, individualism is low on the group dimension preferring increased individual autonomy and responsibility (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006).

Studies using cultural cognition often compare the risk perceptions of hierarchical individualists, those high on the grid dimension and low on the group dimension, to their reverse, egalitarian communitarians who are low on grid and high on the group dimension. The differences in risk perception between these pairings are significant for risks related to the environment, abortion, and guns (Kahan et al. 2007). For environmental risks specifically, communitarianism, with its concern about collective well-being and egalitarianism, with its concern about the negative impacts of concentrated power tend to see threats to the natural world as a large risk. Hierarchs believe that natural resources can be well managed by experts and individualism supports the use of natural resources in private enterprise, and therefore hierarchy and individualism tend not to be concerned about environment harm. Studies have shown the differences between hierarchical individualists and egalitarian communitarians on climate change (Kahan et al. 2012) and water supply risk (Overdevest and Christiansen 2013). In terms of the influence of cultural cognition and pro-environmental behavior, the results are somewhat mixed. One study found that hierarchy and individualism predicted fewer self-reported pro-environment behaviors (Markle 2019), whereas another study found no connection between cultural cognition and behavior (Sparks et al. 2021).

As a measure of deep core beliefs, cultural cognition will likely have an indirect association with environmental behaviors, meaning that the relationship will be mediated by environmental orientations. Indeed, cultural cognition is associated with pro-environmental orientation, specifically the New Ecological Paradigm (NEP) (Nowlin 2019; Overdevest and Christiansen 2013).

New Ecological Paradigm

Moving from deep core beliefs to environmental orientations, of first measure of environmental orientations is the New Ecological Paradigm (NEP). The NEP, developed in 1978 (Dunlap and Van Liere 1978), is often used in the VBN framework as a measure of environmental beliefs (Canlas, Karpudewan, and Khan 2022), and has been widely adopted to explain environmental behavior (e.g., Stern 2000). It's original development focused on explaining the values associated with environmentalism as an emergent environmental paradigm

that challenged the dominant social paradigm. Central to the NEP is concern about the environment and beliefs about the proper relationship between humans and the natural world (Stern, Dietz, and Guagnano 1995). The NEP was updated in 2000, and the new measure comprises five aspects which are “the reality of the limits to growth, anti-anthropocentrism, the fragility of earth’s balance, rejection of exemptionism, and the possibility of an ecocrisis” (Dunlap et al. 2000, 432).

In our approach, we view the NEP as an ecological worldview that encompass values associated with the environment, and, like cultural cognition works to ensure that an individual’s beliefs about the environment remain congruent with their ecological worldview. As a worldview, the NEP measures the environmental cognition that shapes how the natural world is seen (Ballew et al. 2019; Henry and Dietz 2012; Xiao, Dunlap, and Hong 2019). Similar to how the NEP has been used in the VBN framework, we consider it an environmental orientation that is situated between values and behaviors; however, we argue that, as a worldview, it is a cognitive component of environmental orientations. As a cognition, it influences thinking about environmental issues, shapes other aspects of environmental orientations, and is associated with pro-environmental behaviors (Derdowski et al. 2020).

There are some significant critiques of the NEP. One study applying scale analysis of the NEP based on a random sample of Norwegians found that the scale is not unidimensional as is often claimed (Grendstad 1999). In another study, Sparks, Ehret, and Brick (2022) criticize the NEP for failing to adequately identify political conservatives who also hold a pro-environment orientation. Most important for our purposes is a study that found the NEP to be a relatively weak predictor of self-reported pro-environmental behavior, accounting for 12% of the variation, when compared in the same sample to the Connectedness to Nature Scale and environmentalist identity (Sparks et al. 2021). This may be because endorsement of the NEP has decreased over time (Lou et al. 2022), or it may be that feeling close and connected to nature is more proximate to behavior than more abstract, cognitive worldviews like cultural cognition and the NEP.

Connectedness to Nature

Our second instrument measuring environmental orientation is the Connectedness to Nature Scale (CNS) (Mayer and Frantz 2004). The CNS seeks to measure an individual’s emotional connection to nature, rather than a cognitive environmental worldview. The CNS is based on the notion that to address environmental problems, people must feel emotionally connected to the non-human community on which we are dependent upon. The idea is that current ecological crises are a direct result of humanity losing touch with the broader community in which we live. As we argue, the NEP captures a cognitive element of environmental orientations and feeling connected to nature capture an affective element.

Research on connectedness to nature consistently suggests that people who feel more psychologically linked to the natural world are more likely to engage in pro-environmental behavior. Two meta-analyses found strong and consistent positive relationships between pro-environmental behavior and a connection to nature (Mackay and Schmitt 2019; Whitburn, Linklater, and Abrahamse 2020). In addition, a psychological connection to nature predicts environmental attitudes and behaviors beyond broader environmental worldview measures, suggesting that connectedness to nature is not just a re-labeling of general environmental concern but a distinct predictor (Nisbet, Zelenski, and Murphy 2009). More recent work extends this finding across

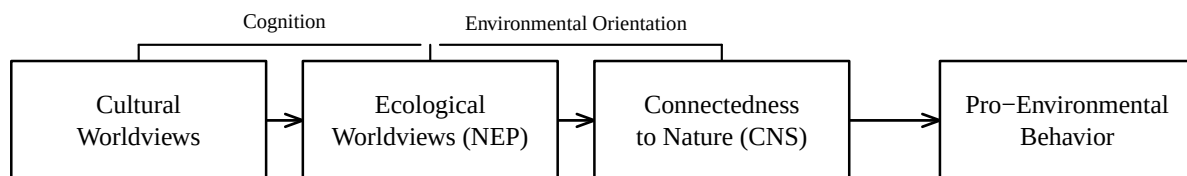
multiple behavioral domains, including civic action, recycling, household conservation, and consumer choices, were all predicted by connectedness to nature, whereas policy support and transportation were predicted by environmental concern, which included the NEP (Gkargkavouzi, Halkos, and Matsiori 2019). One study assessed the role of personality traits on environmental behavior, and provided a comparison of the mediation effect of both the NEP and the CNS on environmental behaviors (Brick and Lewis 2016). The personality traits Openness and Conscientiousness were both shown to be positive predictors of environmental behavior. In both cases, the CNS had a stronger mediation effect than NEP. In other words, the CNS better predicts pro-environmental outcomes.¹ Finally, the CNS (plus control variables) was found to account for 35% of the variance in environmental behaviors (Sparks et al. 2021).

Conceptual Model and Hypotheses

Cultural cognition can be thought of as a deep core belief, and therefore it is an antecedent to environmental orientations. Cultural-based views of risk and the social-psychological tendencies to protect those views likely precede the development of environment-specific cognition, attitudes, and orientations. And given the mixed evidence, is it unclear if the connection between cultural cognition and environmental behavior it is a direct or indirect relationship.

Taking the step from cultural worldviews to environmental orientations, we posit that environmental orientations contain both a cognitive, worldview element that can be measured by the New Ecological Paradigm (NEP) and an affective, emotional connection to nature, as measured by the Connectedness to Nature Scale (CNS). As noted, cultural worldviews are an antecedent to environmental orientations, and similarly, we posit that ecological worldviews are an antecedent to connectedness to nature. Finally, environmental orientations are predictive of pro-environmental behaviors.² Figure 1 illustrates our conceptual model connecting cognition, based on cultural and ecological worldviews; environmental orientations, including both the ecological worldview component and the affective connection to nature component; and pro-environmental behaviors.

Figure 1: Conceptual model tested with the structural equation model: cultural cognition shapes environmental orientation—first the New Ecological Paradigm (NEP) and then the Connectedness to Nature Scale (CNS)—which in turn shapes pro-environmental behavior.



¹However, these authors also note that the CNS contains language that is more consistent with liberal environmentalism, and so may fail to adequately capture the full dimensionality of environmental support across the political spectrum (Brick and Lewis 2016). Indeed, like the NEP, the CNS is composed mostly of items that tap into politically progressive environmentalism and thus may miss conservative pro-environmental orientation (Sparks, Ehret, and Brick 2022).

²Our expectations and analyses are based on pro-environmental behaviors, overall. We do not distinguish between private or public behaviors.

Based on the model shown in Figure 1, we posit several hypotheses regarding the relationships between cultural cognition, the NEP, the CNS, and environmental behavior. Our hypotheses include the sequence approach shown in Figure 1, where cultural worldviews influence ecological worldviews, ecological worldviews influence the connectedness to nature, and connectedness to nature influence pro-environmental behaviors. We also have direct hypotheses, where cultural and ecological worldviews are directly associated with behaviors. We expect that:

H1: Hierarchical-individualists are associated with a *decrease* in the ecological worldview element of environmental orientation, a *decrease* in the connected to nature element of environmental orientation, and *fewer* pro-environmental behaviors

H2: Egalitarian-communitarians are associated with an *increase* in the ecological worldview element of environmental orientation, an *increase* in the connected to nature element of environmental orientation, and *more* pro-environmental behaviors

H3: Ecological worldviews are associated with an *increase* in connectedness to nature

H4: Both the cognitive and affective elements of environmental orientations are associated with *more* pro-environmental behaviors

Data and Methods

To examine the relationship between cultural cognition, environmental orientations, and behavior we draw on a sample of 501 adults living in the US purchased through Survey Sampling Inc. Overall, the sample is largely reflective of the US population in terms of demographics as 51% of respondents are women, 67% are white, and the average respondent is 44 years of age. Additionally, the average person in our sample has attended some college and reports a household income between \$95,001 and 130,000. The survey was administered online in July 2017. The sample size was determined by cost considerations.³ Our analysis was not pre-registered.

Measures

Our dependent variable is a 12-item scale of self-reported pro-environmental behaviors. Questions were taken from the 2000 Gallup Earth Day Poll. Respondents were given a list of several environmental behaviors and asked, “Which of these, if any, have you, yourself, done in the past year?” Possible responses included, “Yes, have done; No, have not done; No opinion.” Yes, responses were coded as a 1 and no/no opinion were coded as 0. For the purposes of analysis, we treat the behavior measures as observed and we added the number of self-reported behaviors together to produce a measure that ranges from 0 (no behaviors) to 12 (all of the behaviors). Table 1 shows the question wording as well as the mean and sd for each self-reported behavior. We also note the 7 public and 5 private behaviors, but we use the single 0-12 scale for analysis.⁴

³IRB approval for the survey was approved by ... Human Subjects Committee on April 3, 2017, protocol number 37-17-0256.

⁴However, the supplemental materials includes analyses that separate public and private, and the results are substantively the same as presented here.

Table 1: Self-reported pro-environmental behaviors by type (public/private), with means and standard deviations

Behavior	Type	Mean	SD
Been active in a group/organization that works to protect the environment	Public	0.19	0.39
Voted for or worked for candidates because of their environmental positions	Public	0.40	0.49
Contributed money to an environmental, conservation, or wildlife group	Public	0.33	0.47
Contacted a public official about an environmental issue	Public	0.19	0.39
Contacted a business to complain about environmentally harmful products/policies	Public	0.16	0.36
Signed a petition supporting environmental protection	Public	0.36	0.48
Attended a meeting concerning the environment	Public	0.17	0.37
Public behaviors (0-7)		1.80	1.92
Avoided using products that harm the environment	Private	0.66	0.47
Tried to use less water in your household	Private	0.75	0.43
Bought a product because it was better for the environment	Private	0.62	0.49
Voluntarily recycled newspapers, glass, aluminum, motor oil, etc.	Private	0.79	0.41
Reduced your household's use of energy	Private	0.78	0.42
Private behaviors (0-5)		3.60	1.58
Total behaviors (0-12)		5.40	2.86

As shown in Table 1, the mean number of self-reported pro-environmental behaviors was 5.4, with a standard deviation of 2.86 in the 0 to 12 scale. For public behaviors the mean was 1.8, with a standard deviation of 1.92, and the mean number of self-reported private behaviors was 3.6, with a standard deviation of 1.58. Overall, respondents reported far more private than public behaviors. In terms of the behavior themselves, recycling had the highest mean at 0.79, which indicates that 79% of the respondents reported that they recycle, followed closely by reducing household energy use at 0.78; buying a product because it was better for the environment had the lowest mean among private behaviors at 0.62. Among public behaviors, voting for or working for a candidate because of their environmental positions had the highest mean at 0.4, followed by signing a petition supporting environmental protection at 0.36; contacting a business to complain about environmentally harmful products/policies had the lowest mean among public behaviors at 0.16.

Our independent variables are composite measures of cultural worldviews, ecological worldviews (New Ecological Paradigm), and the Connectedness to Nature Scale. For cultural cognition, we rely on the short form of the cultural cognition survey questions which includes three questions for each of the four cultural worldviews

(see Kahan 2012). Respondents were asked their agreement on a 1 (strongly disagree) to 4 (strongly agree) scale with three statements for each of cultural worldviews. To measure the New Ecological Paradigm (NEP) we used agreement with three statements on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree). Finally, for the Connectedness to Nature Scale (CNS) using respondent agreement on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree) with three statements. For both the NEP and the CNS, a higher value indicates a stronger orientation towards the environment.

Table 2 shows the question wording for each measurement item used for the latent constructs as well as the fit statistics for each of the constructs including Cronbach's α , Composite Reliability (CR), and the Average Variance Extracted (AVE). As shown in Table 2, each of the constructs show sufficient reliability (α and CR >0.70) except for the communitarianism scale. However, CR values between 0.60 and 0.70 are acceptable for exploratory analysis (Hair et al. 2015).

Table 2: Measurement items and construct reliability (Cronbach's α , composite reliability, average variance extracted)

Construct	Statement	α	CR	AVE
Hierarchy	We have gone too far in pushing equal rights in this country.	0.83	0.83	0.62
	It seems like blacks, women, homosexuals and other groups don't want equal rights, they want special rights just for them.			
	The women's rights movement has gone too far.			
Egalitarianism	We need to dramatically reduce inequalities between the rich and poor, whites and people of color, and men and women.	0.77	0.79	0.55
	Our society would be better off if the distribution of wealth was more equal.			
	We live in a sexist society that is fundamentally set up to discriminate against women.			
Individualism	If the government spent less time trying to fix everyone's problems, we'd all be a lot better off.	0.80	0.80	0.57
	Government regulations are almost always a waste of everyone's time and money.			
	Our government tries to do too many things for too many people. We should just let people take care of themselves.			
Communitarianism	Sometimes government needs to make laws that keep people from hurting themselves.	0.61	0.64	0.38
	Government should put limits on the choices individuals can make so they don't get in the way of what's good for society.			

Construct	Statement	α	CR	AVE
CNS	The government should do more to advance society's goals, even if that means limiting the freedom and choices of individuals.	0.76	0.72	0.47
	I think of the natural world as a community to which I belong.			
	I often feel a kinship with animals and plants.			
NEP	I feel as though I belong to the Earth as equally as it belongs to me.	0.70	0.73	0.48
	Humans are severely abusing the environment.			
	The so-called "ecological crisis" facing humankind has been greatly exaggerated (reverse-coded).			
	If things continue on their present course, we will soon experience a major ecological catastrophe.			

On the cultural worldview measures, we combined the scales to create dummy variables for hierarchical-individualists and egalitarian-communitarians. Respondents who scored at or above the median for both scales were coded as 1, all others were coded as 0 (similar to Kahan et al. 2012). Overall, 213 respondents (42.5%) were classified as hierarchical-individualists, and 176 respondents (35.1%) were classified as egalitarian-communitarians.

To test our hypothesis, we estimated a Structural Equation Model (SEM) using the lavaan package in R (Rosseel 2012). The dependent variable, self-reported pro-environmental behaviors, is an observed variable and the cultural cognition worldviews, the CNS, and the NEP scale are latent constructs using the measures in Table 2. In addition, we examine the mediation effects of environmental orientation by examining the direct, indirect, and total effects of the cultural worldviews on pro-environmental behavior.⁵

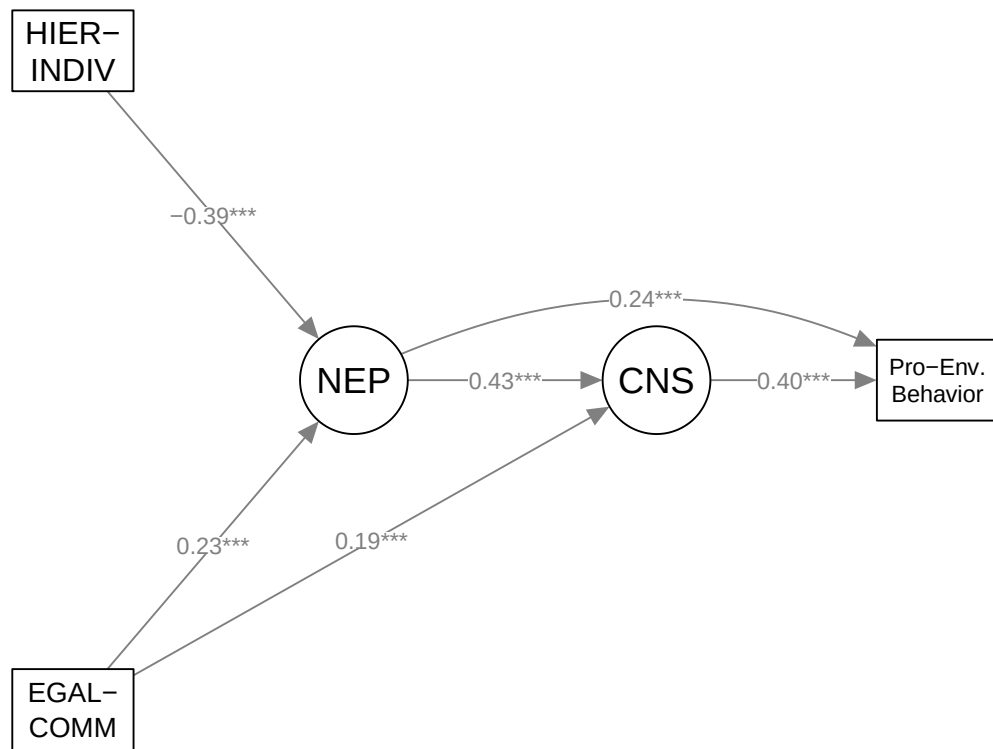
Results

The results for the SEM are shown in Figure 2. For clarity we only include the structural model and only the paths that were statistically significant at $p < 0.05$.

As can be seen in Figure 2, as expected, hierarchical-individualists were negatively associated with the ecological worldviews (NEP). This is as expected because a positive value indicates a more environment-centric worldviews. Also as expected, egalitarian-communitarians were positively associated with the NEP. Additionally, the cognitive element of environmental orientations, the NEP, was positively associated with the affective element, the CNS, as expected. Regarding pro-environmental behaviors, as expected both elements of environmental

⁵In the analysis we control for demographic variables, including age, gender (male=1), race (white=1), education, and income. We chose white males as the indicator variables because of the well-known *white-male effect* associated with risk perceptions and cultural cognition (e.g., Kahan et al. 2007). More information about the demographic variables is available in the supplemental materials.

Figure 2: Structural model of cultural worldviews, ecological worldviews, connectedness to nature, and pro-environmental behaviors.



Note: Cultural worldview groups (HIER-INDIV, EGAL-COMM) → New Ecological Paradigm → Connectedness to Nature → pro-environmental behavior (0–12 scale, public and private behaviors combined), with direct paths. Controlling for age, gender, race, education, and household income (paths not shown). The measurement (indicator) portion is omitted for clarity; standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The full path diagram including the measurement model, and the equivalent model with the four separate worldview measures, are in the supplemental materials.

orientations were positively associated with an increase in the number behaviors; however, there was no direct relationship to cultural worldviews and behavior. Finally, egalitarian-communitarians were more likely to feel connected to nature, therefore, they had the strongest environmental orientation as both the cognitive and affective elements were significant.

The path coefficients are standardized, which means that they equal the expected change in the standard deviation (sd) of the outcome variables for each standard deviation of the predictor, keeping all other variables constant. For example, the path coefficient from CNS to behaviors is 0.404, so holding the other variables constant, a 1-sd increase in the CNS scale is associated with a 0.404-sd increase in the number of behaviors. In terms of the number of actual predicted behaviors, as shown in Table 1, the sd of behaviors is 2.86 and 0.404×2.86 equals 1.155 predicted additional behaviors for each 1-sd increase in the CNS.

Standardized path coefficients allow for a comparison of strength of the various relationships. As shown, the strongest relationship is between the two environmental orientation measures, with the NEP to CNS path = 0.425. In terms of the relationship to behaviors, the CNS path is the strongest at 0.404 followed by the NEP to behavior path at 0.240. No other paths to behaviors is significant. For cultural worldviews to environmental orientations, hierarchal individualists have the strongest (negative) relationship, at -0.392, with ecological worldviews. Egalitarian communitarians have a stronger relationship to ecological worldviews at 0.231 than connectedness to nature at 0.188, but are significantly associated with both.

The fit statistics for the model are shown in Table 3, and include the Root Mean Square Error of Approximation (RMSEA), the Comparative Fit Index (CFI), the Tucker-Lewis Index (TLI), the Root Mean Square Error of Approximation (RMSEA) and its confidence intervals, the Standardized Root Mean Square Residual (SRMR), and the AIC and BIC. Overall, the statistics indicate a reasonable fit for the model (Hu and Bentler 1999).

Table 3: Model fit statistics for the structural model

Model	Chi-sq	df	CFI	TLI	RMSEA	RMSEA 90% CI	SRMR	AIC	BIC
Full (HI/EC, + direct paths)	141.46	40	0.905	0.834	0.071	(0.059, 0.084)	0.039	10,559.8	10,715.9

Note: Model controls for age, gender, race, education, and household income. Conventional benchmarks: CFI/TLI ≥ 0.90 acceptable, ≥ 0.95 good; RMSEA ≤ 0.08 acceptable, ≤ 0.05 good; SRMR ≤ 0.08 good (Hu and Bentler 1999).

Next we examine the direct, indirect, and total effects of cultural worldviews on behaviors. Those results are shown in Table 4.

Table 4: Standardized decomposition of cultural worldview group effects on pro-environmental behavior.

Predictor	Direct	Serial (NEP-CNS)	Total indirect	Total
Hierarch-Individualist	0.036	-0.067***	-0.157***	-0.121**

Predictor	Direct	Serial (NEP-CNS)	Total indirect	Total
Egalitarian-Communitarian	0.028	0.040***	0.171***	0.199***

*Note: Effects are from the full combined model, controlling for age, gender, race, education, and household income. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.*

For hierarchical individualists there was a total effect on behaviors of -0.121 and for egalitarian communitarians it was 0.199. However, as shown in Figure 2 and Table 4, there is no direct relationship from cultural worldviews to behaviors, therefore the effects are strictly a result of indirect effects as environmental orientations appear to mediate the relationship between cultural worldviews and behaviors.

The serial effect includes the paths of the cultural worldviews to the NEP, NEP to CNS, and CNS to behaviors. So, for example, for hierarchical individualists the serial effect is calculated by $-0.392 \times 0.425 \times 0.404 = -0.067$. The total indirect effect captures the paths from cultural worldviews to NEP, NEP to behaviors, cultural worldviews to CNS, and the serial effect. For hierarchical individualists that is $(-0.392 \times 0.240) + (0.010 \times 0.404) + (-0.067) = -0.094 + 0.004 - 0.067 = -0.157$. Finally, the total effect is simply the direct effect plus the total indirect effect.

Discussion

Scholars have noted a gap between values and beliefs about the environment and pro-environmental behavior. In particular, the cultural worldviews derived from cultural cognition, specifically egalitarianism and communitarianism, have been shown to inform views about several environment issues as well as overall attitude toward the environment. However, little research has found a connection between cultural worldviews and environmental behavior (but see Markle 2019). In this paper, we developed and tested a conceptual model, based on the existing VBN and ACF frameworks, where cultural worldviews influence environmental orientations, and environmental orientations influence behaviors. Additionally, our contribution notes that cultural worldviews and ecological worldviews are cognitive features of belief systems that provide both a basis of values and beliefs as well as a series of psychological mechanisms that ensure that beliefs and attitudes are consistent with the underlying worldviews. In turn, these cognitive features of belief systems influence the affective element of environmental orientations, which is the emotional connection to nature. Finally, the affective element of environmental orientations is what is most strongly associated with pro-environmental behaviors.

Based on our conceptual model, we posited that egalitarian communitarians would be positively associated with environmental orientations and environmental behavior, and hierarchical individualists would be negatively associated with environmental orientations and environmental behavior. Additionally, we expected that the cognitive element of environmental orientations would be associated with the affective element, and that both elements of environmental orientation would be associated with increased self-reported pro-environmental behaviors. Overall, we found mixed results for our hypotheses. As expected, egalitarian communitarians were associated with both the CNS and NEP; however, they were not directly associated with behaviors.

In addition, hierarchical individualists were less likely to have an ecological worldviews, but they were no more or less likely to feel connected to nature than the other cultural worldviews. Finally, both elements of environmental orientations were associated with more pro-environmental behaviors, and, based on the standardized coefficients, the affective element (CNS) had a stronger relationship than the cognition element (NEP).

Given the uncertainty about the relationship between cultural worldviews and pro-environmental behaviors, we also did a mediation analysis to examine if environmental orientations mediate the relationship between cultural worldviews and behaviors. If so, cultural worldviews would still influence behavior, but only indirectly. We found that both the cognitive and affective elements of environmental orientations mediated the relationship between cultural worldviews and environmental behaviors. This means that the effect of cultural worldviews on pro-environmental behaviors is based solely on their influence in environmental orientations. Along with the results of the structural equation modeling, this helps us understand cultural cognition as an antecedent of environmental orientation and fill some of the attitude-behavior gap.

We also found that the Connectedness to Nature Scale (CNS) had more predictive power regarding environmental behaviors than the NEP based on the standardized path coefficients shown in Figure 2. This finding is in line with work showing that the CNS is a stronger predictor than the NEP (e.g., Sparks et al. 2021). Additionally, it seems that the CNS and NEP are measure two distinct concepts that can be grouped under the umbrella of environmental orientation, where the NEP is explicitly a measure of environmental-attitude while CNS is a measure of emotional connection to nature. Overall, the affective element of environmental orientations is more proximate to behaviors than the cognitive element. Along similar lines, a recent study found that harnessing the right mix of emotions is a powerful motivator of climate action (Kovács et al. 2024).

Another reason that the affective element of environmental orientations, the CNS, displays more power than the cognitive NEP to influence behavior is that there has been a slow decline in environmental concern as measured by the NEP (Lou et al. 2022). The downtrend of the NEP was shown through meta-analysis; however, there was no indication of why this may be happening. We are not aware of a similar study tracking CNS over time, but that may prove a fruitful avenue for further study. If it is only the NEP that is declining over time, that may indicate that the measurement instrument is becoming outdated, potentially because of shifts over time making the distinction between the dominant social paradigm and the new ecological paradigm less apt. For example, many see technological innovation as the key to unlocking a new green economy, rather than destroying the earth. Relatedly, perhaps the language used in the NEP has just become outmoded and out of line with how environmentalists currently would express environmental attitude (see Sparks, Ehret, and Brick 2022 for a discussion of this).

There are a few limitations with this paper. First, we have a relatively small sample size with 501. While we have no reason to think that a larger sample size would substantively change the results, the results should be replicated with more respondents. Additionally, our data is cross-section, and future research should measure the nature of the relationships between cultural cognition, environmental orientations, and environmental behaviors over-time using panel data. Panel data could explore the strength and consistency of the relationships over time. Finally, causality is difficult to determine using SEM on cross-sectional data. While we argue that there are theoretical and empirical reasons to see cognition preceding affective elements of belief systems,

we recognize that the statistical associations we found does not constitute definitive evidence of causality. Therefore, we have tried to be careful in our language describing our findings.

Conclusion

Achieving environmentally sustainable outcomes require a significant engagement of pro-environmental behavior across all segments of society. While a large body of research exists that examines pro-environmental behaviors, some gaps remain. In this paper, we developed a conceptual model that draws a distinction between cognitive and affective elements of belief systems, and tested hypotheses based on that approach. We argue that combining cultural cognition and environmental orientation should improve our understanding of the drivers of environmentally protective behaviors

Because cultural cognition shapes how an individual responds to risk (e.g., Kahan et al. 2012), it was something of a puzzle as to why it only weakly correlates to behavior that protects the environment – an action that can be seen as a way to reduce environmental risk to the individual. The evidence we presented in this paper suggests that environmental orientation – whether attitude (NEP) or emotional connection (CNS) – is an important link in this relationship. Additionally, we presented evidence that cultural worldviews are best thought of as indirect predictors of pro-environmental behavior. The relationship was mediated by environmental orientations as measured by both the NEP and the CNS. Further, by explaining behavior with a mediation model we could close some of the so-called attitude-behavior gap. When cultural cognition and environmental orientation are aligned, people are more likely to engage in environmentally protective behavior.

Another implication of this study is further evidence that CNS is a stronger predictor of behavior than NEP. This may be useful for researchers trying to decide which measure of pro-environmental orientation to include in their questionnaire. We would like to see further testing of this finding in studies in other contexts and through meta-analytic techniques.

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